Alex Tang

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Education

University of California, Los Angeles (UCLA) — Los Angeles, CA
Bachelors of Science in Mathematics of Computation — *2026 – Present*
- **Relevant Courses**: Introduction to Computer Science, Introduction to Discrete Structures, Linear Algebra

Amador Valley High School — Pleasanton, CA
Valedictorian (1/646) | Unweighted GPA: 3.98 | Weighted GPA: 4.70 — *2022 – 2026*

Technical Skills

Computer Languages: Java, Python, C++, R, MATLAB
Tools: OR-Tools, Gurobi, ChromaDB, Ollama, Jupyter, MySQL

Work & Research Experience

Complex Systems Research (SRA, UC Santa Barbara) | *MATLAB, Simulink* — 2025
- Applied dynamical systems, control theory, and optimization to model complex biological and engineered systems
- Modeled physical effects of network delay and packet loss on 3-DOF robotic arm operational accuracy
- Co-authored a research paper detailing the degradation thresholds of network latency

Mathematics Tutor — 2024 – 2026
Kumon & Afficient Academy
- Tutored over 70 students in advanced mathematics curricula up to calculus

Mathematical Modeling of Biological Systems (COSMOS, UC Davis) | *R* — 2024
- Created mathematical models using ODEs to simulate scenarios in probability, game theory, and network theory
- Applied ideas to real-life phenomena in epidemiology, ecology, and physiology
- Developed a mathematical model simulating biomass diffusion across a channel-like landscape

Technical Projects

Independent Project: Local RAG System | *Python, ChromaDB, Ollama, Jupyter* — 2026
- Built a local Retrieval-Augmented Generation (RAG) system with integrated document embedding and querying
- Deployed open-source LLMs like qwen3 for local data processing without relying on external APIs

Independent Research Project: Anti-Poaching Model | *Python, OR-Tools, Gurobi, MySQL* — 2023 – 2026
- Applied linear and integer programming to route park rangers through Hell's Gate National Park hotspots
- Implemented a MySQL database to efficiently query and store mathematical model data constraints

Co-Author, NeuroPass: Advanced Precision Therapy for MS — 2025
- Designed an AI and ultrasound-guided quantum dot drug delivery concept for Multiple Sclerosis treatment
- Recognized as Co-Author, earning a Toshiba ExploraVision Honorable Mention (top 10% of 15,000)

Leadership & Extracurriculars

Bruin Software Engineers (BSE) | *Health Tech in Hardware Track* — 2026 – Present
- Collaborating on hardware-integrated health technology solutions within a student engineering team
- Bridging physical sensors and hardware components with software-driven data processing pipelines

Competitive Programmer — 2024 – Present
Codeforces & USACO
- Achieved Codeforces Rating of 1726 (Expert tier)
- Promoted to USACO Gold division (2025)
- American Invitational Math Exam (AIME) Qualifier: 121.5 on AMC 10A (Top 5%)

Co-President — 2024 – 2025
Music4America (M4A) | Member (2019 – 2026)
- Organized community events raising over $5000 for local school district music programs
- Scaled non-profit operations, nearly doubling membership from 127 to 244 members
- Coordinated and performed in 30+ bi-weekly musical recitals at local senior centers and VA hospitals
- Instructed mathematics camps for middle school students, generating over $1000 in fundraising revenue